

BeeBop

Development Plan

Version 2.0 | Conversational Property Marketplace | Nigeria

Document Type	Development Plan — Version 3.0
Supersedes	Development Plan Version 2.0 (April 2026)
Classification	Confidential
Platform	Web Application (PWA for Inspector Portal)
Target Market	Abuja, Nigeria — initial rollout
Total Timeline	12 months from project kick-off to full production launch

Version 2.0 — Change Log

This version incorporates all product design revisions agreed following the v1.0 review. The changes below affect the fee structure, technology stack, notification channels, onboarding flow, AI/NLP architecture, result ranking logic, valuation report scope, conversation memory model, and the addition of short-let in-booking messaging.

Area	Change	Sections Affected
Listing fee	Removed entirely. All listings are free to upload. BeeBop earns only at transaction completion.	Fee Structure, Phase 1 Sprint 1
Rent fee structure	Revised to two tiers: up to 5m — 20k flat + 3% annual rent (landlord 15k + 2%, seeker 5k + 1%). Above 5m — 20k flat + 2% annual rent. No cap.	Fee Structure, Phase 2 Sprint 10
Student accommodation fee	2.5% from accommodation owner + 3% from seeker. Charged independently per party.	Fee Structure, Phase 2 Sprint 10
Short-let fee	Host 3% always. Seeker 10% on bookings up to 150k, 8% above 150k.	Fee Structure, Phase 2 Sprint 11
Sales fee	3% up to 50m sale value, 2.5% above. Paid entirely by seller. Invoice at signing; agreement withheld until payment confirmed.	Fee Structure, Phase 2 Sprint 10
SMS notifications and verification	Removed. Replaced with email (Resend) and WhatsApp Business API. Twilio and Termii removed from stack.	Tech Stack, All notification sprints
Backend language	Python replaces Node.js for all backend services. FastAPI, SQLAlchemy, Alembic, Celery, Pydantic replace NestJS, Prisma, BullMQ.	Tech Stack, All backend sprints
Seeker onboarding — area dropdown	Removed from Step 4. Onboarding now collects category preference only (plus institution, level, gender for students).	Phase 1 Sprint 1
Conversation memory	Session-only. No persistent chat history. No Chats sidebar section. Session context in Redis expires on session end or 30 minutes inactivity.	AI/NLP, Phase 3 Sprint 12
Result ranking	Saves and enquiries removed as ranking signals. Ratings weighted as second signal after verification status.	AI/NLP, Phase 3 Sprint 12
Valuation report	Fair market value indicator and comparable listings removed. Report now contains: area infrastructure scores, inspector professional note, report date only.	Phase 2 Sprint 7

Short-let messaging	In-booking group chat added. Seeker, host, and BeeBop moderator. Content filtering as secondary safety layer. On-platform action shortcuts within chat.	Phase 2 new sprint
Sales fee payer	Clarified: seller pays 100% of sales commission. Buyer has zero platform fees.	Fee Structure
Rent flat fee split	Clarified: landlord pays 15k, seeker pays 5k of the 20k flat component.	Fee Structure
Hosting — zero cost MVP	AWS replaced by free-tier services: Vercel (frontend), Render free tier (backend with keepalive), Neon (PostgreSQL), Upstash (Redis), Cloudinary (media). All free at MVP scale.	Tech Stack, Infrastructure
Render keepalive strategy	UptimeRobot 5-minute pings and GitHub Actions scheduled workflow prevent Render free tier spin-down. /health endpoint added to FastAPI backend from Sprint 1.	Tech Stack, Phase 0, Phase 1 Sprint 1
LLM API accounts pre-funded	Claude (Anthropic) and GPT-4o (OpenAI) accounts already funded. No cost concern for AI services during MVP.	Tech Stack, Prerequisites
Domain confirmed	beebop.ng domain purchase confirmed by founder. Treated as the sole unavoidable external cost.	Prerequisites

1. Introduction

This document defines the complete development plan for BeeBop — a conversational, fully intermediated property marketplace for the Nigerian real estate market with initial operations in Abuja. Version 3.0 supersedes the April 2026 v2.0 plan and incorporates the final infrastructure and cost decisions for a zero-cost MVP approach.

The plan covers team structure and roles, the full technology stack, a phased twelve-month delivery roadmap, sprint-level task breakdowns, quality assurance strategy, infrastructure and deployment, and risk management. It is the primary reference document for the development team.

Planning Principle

BeeBop is built in phases, each delivering a usable and revenue-generating increment before the next begins. No phase waits for a perfect system before shipping. The goal is to reach real users with real listings as early as possible and iterate from a live foundation.

2. Team Structure and Roles

2.1 Core Build Team

Role	Responsibility	Phase Active	FTE
Technical Lead / Architect	Overall architecture. API design. Database schema. Integration strategy for LLM, WhatsApp, payments, and identity. Code review authority.	All phases	1
Backend Developer (Python)	FastAPI services. SQLAlchemy models. Alembic migrations. Authentication. Payment integration. Celery task definitions. Admin and inspector portal backend.	All phases	1–2
Frontend Developer	Main page, conversational UI, category browse, listing page, all dashboards, internal staff portal frontend. Next.js throughout.	Phase 1 onwards	1–2
AI / NLP Engineer	Claude and GPT-4o integration. Prompt engineering and registry. Intent classification pipeline. Parameter extraction. Session context management. Query logging.	Phase 2 onwards	1
UI / UX Designer	Wireframes and component design. Design system. Responsive layouts. Listing card, gallery, dashboard, and CTA bar components.	Phase 0–2	1
QA Engineer	Test plan authoring. Manual and automated test execution. Regression testing. UAT coordination.	Phase 1 onwards	1

2.2 Specialist Contributors

Role	Responsibility	Engagement
Property / Legal Consultant	Finalise tenancy and sale agreement templates. Advise on Nigerian tenancy law compliance. WhatsApp Business API terms review. NDPR compliance.	Retained — Phases 1–2
NIMC Integration Specialist	Configure and test real-time NIN verification via NIMC API.	Contract — Phase 1
WhatsApp Business API Specialist	Configure Meta Business Manager, WhatsApp Business API access, OTP and notification message templates. Manage template approval process.	Contract — Phase 0–1

DevOps / Infrastructure Engineer	AWS provisioning. CI/CD pipeline. Environment management. Monitoring and alerting. Backup and recovery.	Contract — Phase 2 onwards
Security Consultant	Security audit of auth flows, API endpoints, file uploads, and data handling before production launch.	Contract — Phase 3

2.3 Operational Roles (Pre-Launch)

Role	Responsibility	Start
Platform Admin (initial)	Reviews document submissions, approves inspection reports, manages verification queue, assigns visits, moderates short-let group chats.	Phase 2 — before beta
Lead Inspector	First inspector. Establishes field protocol, calibrates checklist, trains subsequent inspectors.	Phase 2
Trusted Agent(s)	First agents onboarded before beta. Attend property visits as BeeBop representatives.	Phase 3
Short-Let Chat Moderator	Monitors short-let in-booking group chats. Reviews flagged messages. Can be the same person as platform admin at low volume.	Phase 2 — before beta
Customer Support	Handles seeker and landlord queries. Escalates to admin.	Phase 3

3. Technology Stack

Stack Principle

Python owns all backend services. JavaScript/TypeScript owns all browser interfaces. This separation is clean, enforced at the repo level, and not crossed.

3.1 Frontend

Layer	Technology	Rationale
Framework	Next.js (React)	SSR for listing page SEO. App router for clean page structure. PWA support for inspector portal.
Styling	Tailwind CSS	Rapid UI development. Consistent design tokens. Well-suited to dashboard-heavy interfaces.
State management	Zustand	Lightweight. Handles session state, filter state, and short-let booking cart state without Redux overhead.
Maps	Google Maps JavaScript API	Listing location display, area pin views, proximity calculations.
Date picker	React Day Picker	Short-let availability calendar and rent move-in date filter.
PWA (inspector)	next-pwa	Service worker and offline caching. Background sync for queued inspector submissions.
Forms	React Hook Form	Form handling and client-side validation across onboarding, listing creation, and inspection forms.
Real-time chat	WebSocket (native browser API)	Short-let in-booking group chat. Also used for in-app notification delivery.

3.2 Backend (Python)

Layer	Technology	Rationale
Language	Python 3.12+	Primary backend language throughout. Strong AI/ML ecosystem. Developer availability in Nigeria and globally.
Framework	FastAPI	Async-first. Automatic OpenAPI documentation. Native Pydantic integration. High performance. Industry-standard for Python APIs in 2025–2026.
ORM	SQLAlchemy 2.0	Mature, battle-tested Python ORM. Async support. Well-documented. Works cleanly with FastAPI.

Migrations	Alembic	Schema migration management. Integrates directly with SQLAlchemy. Tracks all schema changes as versioned scripts.
Data validation	Pydantic v2	Request and response model validation. LLM output schema validation. Native to FastAPI — no additional library needed.
Database	PostgreSQL	Relational integrity for listings, users, offers, agreements, roles. JSONB fields for flexible amenity and area score data.
Background tasks	Celery + Redis	Notification dispatch, agreement generation, badge award jobs, offer expiry timers, scheduled renewal prompts. Redis as message broker and result backend.
Caching	Redis	Session context (current conversation, search parameters). Valuation report caching. Frequently read listing data.
File storage	AWS S3 + Cloudinary	S3 for inspector evidence, agreement PDFs, raw media. Cloudinary for optimised listing photo delivery.
Authentication	JWT + OTP (email / WhatsApp)	Stateless JWT. Short-lived access tokens with refresh rotation. OTP delivered via Resend (email) or WhatsApp Business API.
API documentation	FastAPI auto-generated OpenAPI / Swagger	Available in development and staging. Disabled in production.
Search	PostgreSQL FTS (Phase 1–2), Elasticsearch (Phase 3+)	PostgreSQL FTS sufficient for MVP. Migrate when listing count exceeds 2,000 or query latency degrades.

3.3 AI and NLP

Component	Technology	Detail
Primary LLM	Anthropic Claude API — claude-sonnet-4-6	Query parsing, parameter extraction, valuation report generation. Selected for context window size and structured output reliability.
Fallback LLM	OpenAI GPT-4o API	Vendor resilience. Image analysis tasks.
LLM SDK	anthropic Python SDK	Official Anthropic Python library. Used directly — no LangChain or similar abstraction layer added.
Prompt management	Versioned prompt registry (Python module)	All prompt templates stored as versioned Python strings in the codebase. Changes tracked in Git. No live prompt editing.
Output validation	Pydantic v2 models	Every LLM response producing structured data is validated against a Pydantic model. Validation

		failure triggers retry (max 2), then keyword search fallback.
Session context	Redis (session-scoped, TTL 30 minutes)	Current conversation turns, current search parameters, last result set. No persistent cross-session storage. Expires on session end or inactivity.
Query log	PostgreSQL + S3	Structured query metadata (intent, extracted parameters, result IDs, click-through) in PostgreSQL. Raw turns in S3 for cost-efficient retention. No personally identifiable conversation content stored.

3.4 External Integrations

Service	Provider	Purpose
Identity verification	NIMC API	Real-time NIN validation. Landlord and agent onboarding.
Business verification	CAC API	Agency CAC registration number validation.
Email OTP and transactional email	Resend	Verification OTPs, agreement PDFs, fee receipts, onboarding confirmation emails. Primary email delivery service.
WhatsApp OTP and notifications	WhatsApp Business API (via Meta or BSP)	Verification OTPs for users preferring WhatsApp. Time-sensitive notifications: offer received, agreement ready to sign, badge issued, visit confirmed.
Short-let group chat	Custom WebSocket service (internal)	In-booking group chat between seeker, host, and BeeBop moderator. Built in-house on the platform — not a third-party chat SDK.
Payment processing	Paystack	All payment events: facilitation fees (rent, student, sales invoice), short-let booking payments, host payouts.
Maps and geocoding	Google Maps Platform	Location display, pin map, landmark proximity, area score geography layer.
Cloud infrastructure	AWS	EC2/ECS for application. RDS for PostgreSQL. S3 for file storage. CloudFront CDN. Route 53. Secrets Manager.
Error monitoring	Sentry	Real-time error tracking. Both frontend (Next.js) and backend (FastAPI) instrumented.
Product analytics	PostHog	Funnel tracking, event logging, feature flags for staged rollout.

3.5 Infrastructure — Zero Cost MVP Stack

All hosting infrastructure uses free tiers for the MVP. Every service listed below has a clear upgrade path when revenue justifies it. No AWS account is needed at this stage.

Component	Platform	Free Tier Detail	Upgrade Path
Frontend hosting	Vercel — Hobby plan	Unlimited deployments. Global CDN. Custom domain. SSL included. No usage limits for a small team.	Vercel Pro at \$20/month when needed.
Backend hosting	Render — Free tier	750 hours/month of compute. Spins down after 15 minutes of inactivity. Keepalive strategy eliminates this (see 3.6).	Render Starter at \$7/month removes spin-down permanently.
Database	Neon — Free tier	PostgreSQL. 512MB storage. No inactivity pause. Serverless scaling.	Neon Pro at \$19/month for larger storage and compute.
Redis	Upstash — Free tier	10,000 commands/day. 256MB storage. Serverless. Sufficient for session context and Celery queue at MVP scale.	Upstash Pay-as-you-go at \$0.2 per 100,000 commands — effectively free until significant scale.
Media storage	Cloudinary — Free tier	25GB storage. 25GB bandwidth/month. Image transformations included. Covers all listing photos and inspector media at MVP scale.	Cloudinary Plus at \$89/month when storage approaches 25GB.
Email	Resend — Free tier	3,000 emails/month. 100/day. Covers OTP delivery and formal document emails at MVP scale.	Resend Pro at \$20/month for 50,000 emails/month.
CI/CD	GitHub Actions — Free tier	2,000 minutes/month for private repos. Covers all test and deployment pipelines for a small team.	GitHub Teams at \$4/user/month if minutes are exhausted.
Error monitoring	Sentry — Free tier	5,000 errors/month. 10,000 performance transactions. Sufficient for MVP.	Sentry Team at \$26/month when needed.
Product analytics	PostHog — Free tier	1 million events/month. Will not be exceeded at MVP scale.	PostHog Scale when event volume grows.
Secrets management	Environment variables in Render + Vercel dashboards	Render and Vercel both provide encrypted environment variable storage. No third-party secrets manager needed at MVP stage.	AWS Secrets Manager when infrastructure scales to AWS.

Backup	Neon automated backups	Neon free tier includes point-in-time restore. S3 not needed at this stage — Cloudinary has its own redundancy.	Dedicated backup strategy when moving to paid infrastructure.
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3.6 Render Keepalive Strategy

Render's free tier spins down a service after 15 minutes of no incoming HTTP traffic. Two complementary mechanisms prevent this from affecting users.

Mechanism	Setup	How it works
UptimeRobot — primary	Create free account at uptimerobot.com . Add monitor pointing to <code>https://[your-render-url]/health</code> . Set interval to 5 minutes.	Sends an HTTP GET to the health endpoint every 5 minutes. Render never reaches the 15-minute inactivity threshold. Also serves as genuine uptime monitoring — alerts the team if the backend actually goes down.
GitHub Actions scheduled workflow — backup	Add a <code>.github/workflows/keepalive.yml</code> file to the repository. Schedule: <code>cron every 14 minutes</code> . Step: <code>curl the /health endpoint</code> .	Acts as a secondary pinger if UptimeRobot has any delay. Lives in version control alongside the codebase. Zero external account needed beyond GitHub.
/health endpoint — required from Sprint 1	Add a single GET <code>/health</code> route to the FastAPI app returning <code>{status: ok}</code> with HTTP 200.	Lightweight endpoint that both pingers hit. Also used by monitoring tools and future load balancers. Must respond in under 100ms.

When to upgrade Render

Move to Render Starter (\$7/month) before the beta launch in Phase 3. At that point real users are on the platform and cold starts — even rare ones — are unacceptable. The keepalive strategy is a development and pre-beta measure, not a permanent production solution.

4. Fee Structure

All listings are free to upload. BeeBop earns exclusively at the point of transaction completion. There are no upfront listing fees, no document submission fees, and no inspection fees charged to listers.

4.1 Rent

Tier	Annual Rent	Landlord Pays	Seeker Pays	Total BeeBop Revenue
Standard	Up to 5,000,000 Naira	15,000 Naira flat + 2% of annual rent	5,000 Naira flat + 1% of annual rent	20,000 Naira + 3% of annual rent
Premium	Above 5,000,000 Naira	15,000 Naira flat + approx. 1.34% of annual rent	5,000 Naira flat + approx. 0.66% of annual rent	20,000 Naira + 2% of annual rent

Rent Fee Notes

Flat fee split: landlord always pays 15,000 Naira, seeker always pays 5,000 Naira of the flat component, regardless of tier. The percentage components maintain approximately a 2:1 landlord-to-seeker ratio. No cap applies at any rent level. All fees collected at agreement signing via Paystack.

4.2 Student Accommodation

Party	Fee	Timing
Accommodation owner	2.5% of agreed accommodation price	Collected at agreement signing via Paystack
Seeker (student)	3% of agreed accommodation price	Collected at agreement signing via Paystack
Total BeeBop revenue	5.5% of agreed accommodation price	Both charges are independent — not pooled

4.3 Short-Let

Tier	Booking Value	Host Pays	Seeker Pays	Total BeeBop Revenue
Standard	Up to 150,000 Naira	3% of booking value	10% of booking value	13% of booking value
Premium	Above 150,000 Naira	3% of booking value	8% of booking value	11% of booking value

Short-Let Fee Notes

Booking value is the total stay amount (nightly rate x nights), not the per-night rate. Threshold check runs at payment capture time. Host payout is gross booking value minus host fee (3%), disbursed after check-in confirmation. Seeker fee added to booking total at checkout.

4.4 Sales

Tier	Sale Value	Seller Pays	Buyer Pays	Total BeeBop Revenue
Standard	Up to 50,000,000 Naira	3% of sale value	Zero	3% of sale value
Premium	Above 50,000,000 Naira	2.5% of sale value	Zero	2.5% of sale value

Sales Fee Notes

Buyers pay zero platform fees on sales transactions — a strong acquisition message. BeeBop generates a Paystack invoice to the seller at agreement signing. The signed agreement is withheld until invoice payment is confirmed. Seller has 48 hours to pay before the agreement is automatically released to both parties pending manual admin review.

4.5 Renewal Fees

Type	Fee	Paid By	Trigger
Rent renewal	10,000 Naira flat	Landlord	On tenancy renewal. No new inspection. Simplified agreement flow.
Student accommodation renewal	Per-semester or annual fee recalculated at current rate	Both parties	On accommodation renewal at end of agreed term.
Short-let	No renewal fee	N/A	New booking created for each stay.

Sales	No renewal — one-time transaction	N/A	N/A
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5. Development Phases — Overview

The twelve-month programme is divided into five phases. Each phase concludes with a gate review. The gate is a genuine quality check — all milestone criteria must be met before the next phase begins.

Phase	Name	Duration	Milestone
Phase 0	Foundation	Weeks 1–3	Environment, stack, accounts, schema, design system, CI/CD, WhatsApp Business API provisioning.
Phase 1	Core Platform MVP	Weeks 4–14	Register, create listings (free), browse, listing page, admin doc review portal, all dashboards.
Phase 2	Verification, Transactions and Messaging	Weeks 15–27	Inspector portal (offline PWA), verification pipeline, offer/agreement/payment flow, short-let booking, in-booking group chat.
Phase 3	Beta Launch	Weeks 28–37	AI/NLP layer, 50+ real listings, beta seeker cohorts, security audit, go/no-go decision.
Phase 4	Production Launch and Optimisation	Weeks 38–52	Public launch, rating system, knowledge base, NLP optimisation, scaling.

Gate Review Criteria

Each gate requires: (1) all milestone features passing QA on staging, (2) zero critical or high-severity open bugs, (3) relevant stakeholder sign-off. Timeline extends before quality bar drops.

PHASE 0

Foundation

Weeks 1 — 3

Phase 0 eliminates every setup dependency before build work begins. Time spent in Phase 0 is recovered many times over by uninterrupted delivery in subsequent phases.

6.1 Tasks

Task	Owner	Duration	Notes
GitHub repos and branching strategy	Tech Lead	Day 1	Monorepo with workspace separation: /frontend (Next.js), /backend (FastAPI), /inspector (PWA). Branch strategy: main (production), develop (staging), feature branches.
AWS account and base infrastructure	Tech Lead / DevOps	Days 1–3	VPC, RDS (PostgreSQL), S3 buckets, IAM roles, Secrets Manager, CloudFront. Dev and staging environments provisioned.
FastAPI project initialisation	Backend Dev	Days 1–2	Module structure: auth, users, listings, verification, offers, agreements, notifications, payments, inspector, admin, ai_search, chat. Pydantic models scaffolded for each module.
SQLAlchemy models and Alembic setup	Backend Dev + Tech Lead	Days 2–5	All models defined: users (all roles), listings (all types), rooms, offers, agreements, inspection_reports, area_scores, notifications, badges, bookings, chat_threads, chat_messages. Schema review session before first migration.
Next.js frontend initialisation	Frontend Dev	Days 1–2	App router. Page scaffold: main, category browse, listing page, auth, dashboards (seeker, landlord, student, short-let), internal portal (admin, inspector, agent).
WhatsApp Business API provisioning	Tech Lead + WhatsApp Specialist	Week 1 (initiated)	Create Meta Business Manager account. Apply for WhatsApp Business API access. Submit OTP and notification message templates for approval. Template approval takes 24–72 hours.

Resend account and email domain	Tech Lead	Day 1	Create Resend account. Verify sending domain. Configure DKIM/SPF. Test OTP email template.
Paystack merchant account	Founder	Week 1	Complete KYC. Configure settlement account. Obtain test and live API keys.
NIMC and CAC API applications	Founder / Tech Lead	Week 1	Submit access applications. These have external lead times — initiate immediately.
Anthropic and OpenAI API accounts	Tech Lead	Day 1	Create accounts. Set up billing with budget alerts. Store keys in Secrets Manager.
GitHub Actions CI/CD pipeline	Tech Lead	Days 3–5	PR: ruff lint, mypy type-check, pytest unit tests (Python). Merge to develop: deploy to staging via Render webhook. Manual trigger: promote to production. Keepalive workflow added (see 3.6).
Python tooling configuration	Backend Dev	Day 2	pyproject.toml with ruff (linting), mypy (type checking), pytest (testing), black (formatting). Pre-commit hooks configured.
Design system documentation	UI/UX Designer	Weeks 1–2	Colour palette, typography, spacing, component inventory (cards, badges, forms, modals, chat UI). Stored in Figma.
Google Maps Platform setup	Tech Lead	Day 2	Create GCP project. Enable Maps JavaScript API, Geocoding API, Places API. Configure billing with alerts.
Sentry and PostHog setup	Tech Lead	Day 3	Both connected to frontend (Vercel) and backend (Render). Alert routing configured. Feature flag structure defined.
UptimeRobot keepalive setup	Tech Lead	Day 3	Create UptimeRobot free account. Add HTTP monitor pointing to Render backend /health endpoint. Interval: 5 minutes. Alert email configured. Also add GitHub Actions keepalive workflow (.github/workflows/keepalive.yml) as backup pinger.
Phase 1 backlog population	Tech Lead	Week 2–3	All Phase 1 stories written with acceptance criteria and estimates. Sprint 1 ready at Week 4.

6.2 Phase 0 Deliverables

- Operational development, staging, and production infrastructure on AWS.
- Python backend and Next.js frontend projects initialised with CI/CD running.
- SQLAlchemy models and first Alembic migration committed and reviewed.
- WhatsApp Business API access provisioned and OTP template approved.
- All API credentials stored as environment variables in Render and Vercel dashboards.
- Design system documented in Figma.
- Phase 1 sprint backlog fully populated and estimated.
- UptimeRobot keepalive monitor active. GitHub Actions keepalive workflow committed to repo. /health endpoint listed in Sprint 1 acceptance criteria.

PHASE 1

Core Platform MVP

Weeks 4 — 14
(11 weeks / 5–6 sprints)

Phase 1 builds the structural foundation. By the end, seekers can register and browse listings, landlords can register and create free listings, and admins can review documents and issue badges. No payments, no AI search, no transactions — but a complete and navigable platform.

7.1 Sprint 1 — Weeks 4–5: Authentication and User Models

Goal: Users can register via email OTP or WhatsApp OTP, complete onboarding, and reach their appropriate home screen.

Story	Detail	Acceptance Criteria
Seeker registration — email OTP	Email entry screen. OTP dispatched via Resend. OTP verify endpoint. JWT access and refresh tokens issued.	OTP delivered within 30 seconds. Invalid code returns clear error. Resend timer functional.
Seeker registration — WhatsApp OTP alternative	WhatsApp OTP sent via WhatsApp Business API for users who select this option.	WhatsApp message delivered using approved template. Same JWT flow on verification.
Seeker onboarding — basic identity	First name, last name. Email required (used for OTP and formal documents).	Profile record created with correct fields. Cannot proceed without first and last name.
Seeker onboarding — category preference	Single step: What are you looking for? Multi-select: Off-campus, Short-let, Rent, For Sale. No area dropdown. No budget field.	Preferences stored on user profile. User arrives at personalised main page immediately after.
Student seeker variant — Step 4	Institution (searchable, from seeded list). Academic level. Gender. No area, no budget.	Institution and gender stored. Gender silently applied to room-level filtering in search. Level stored for proximity logic.
Landlord registration — email/WhatsApp OTP	Same OTP dual-channel as seeker.	Identical OTP flow. Role stored as landlord on user record.
Landlord — account type selection	Individual or Agency. Routes to different verification path.	Account type stored. Individual goes to NIN verification. Agency goes to CAC verification.
Landlord — individual NIN verification	Full name, DOB, NIN. NIMC API real-time check via Python requests with retry logic.	Valid NIN: verified flag set. Invalid: retry once. Third failure: admin flag

		raised. NIMC API timeout: async verification with manual fallback.
Landlord — agency CAC verification	Business name, CAC number. CAC API call.	Valid CAC: verified. Async fallback with status shown if CAC API queues.
Landlord — profile setup	Profile photo (S3 upload via presigned URL). Bio (optional). Operating area (optional — not required for listing). Agency: logo, display name, size, areas.	Photo stored in S3. Optional fields genuinely skippable with no downstream penalty.
Session management	JWT refresh. Persistent session across page reloads. Logout clears both tokens.	Expired access token auto-refreshes. Logout clears tokens from client. Session survives hard reload.
/health endpoint	GET /health route on FastAPI returning {status: ok} with HTTP 200. Required for UptimeRobot and GitHub Actions keepalive pinger.	Endpoint responds in under 100ms. Returns 200 with JSON body. Accessible without authentication.
Role-based route guards	FastAPI dependency injection for auth. Role-based access: seeker, landlord, admin, inspector, agent.	Unauthenticated request returns 401. Wrong role returns 403.

7.2 Sprint 2 — Weeks 6–7: Listing Creation (All Types, Free)

Goal: Landlords can create and submit listings across all four categories at no cost. All listing types supported including student accommodation inventory and short-let pricing.

Story	Detail	Acceptance Criteria
Listing creation — base form	Title, subtitle (optional), description (min 200 chars), property type, listing category, address with GPS pin, price fields appropriate to category. Draft auto-save.	Required fields validated. Draft saved on each field change. GPS pin coordinates stored. No payment required at submission.
Listing creation — amenities	Structured checklist grouped by Power, Water, Security, Internet, Parking, Kitchen, Laundry. Stored as structured JSONB, not free text.	All groups render correctly. Selections persisted on draft save.
Listing creation — photo upload	Multi-photo upload to S3 via presigned URLs. Drag-to-reorder. Section labelling (Living Room, Bedroom, etc.). Cover photo selection.	Photos upload without page reload. Section labels editable. Order persisted. Cover photo used as card thumbnail.

Listing creation — document upload	PDF and image upload for title documents. Multiple files. Type labelling.	Documents stored in private S3 bucket (not publicly accessible). Presigned URL with short expiry for admin access only.
Student accommodation — unit type and room management	Add unit type (name, beds per room, price). Add rooms. Gender tag per room (Male/Female). Self-contain: no gender tag field shown.	Unit types and rooms stored relationally. Gender tag enforced at room level in database constraint.
Short-let — availability and pricing	Base nightly rate, weekend rate, minimum stay, turnaround window, instant booking toggle.	All pricing stored. Turnaround window creates automatic blocked days. Instant booking flag stored as boolean.
Listing submission (free)	Submit sends listing to admin doc review queue. Status: Under doc review. Landlord sees status on listing card. No payment step.	Listing status transitions correctly. Landlord notified in-app and via email/WhatsApp on submission confirmation.
Listing card component	Image, title, badge status, rating (not for sales), location, price with correct unit per category.	Sales card has no rating row. Price unit adapts per listing type. Unverified state visually distinct from verified states.

7.3 Sprint 3 — Weeks 8–9: Seeker Discovery — Browse and Listing Page

Goal: Logged-in seekers can browse all categories with working filters. Unauthenticated visitors can view listing pages (valuation report gated). Map view functional.

Story	Detail	Acceptance Criteria
Main page layout	Sidebar: New Chat, category links (Off-campus, Short-let, Rent, Sales). No Chats history section. Chat input centred with suggestion pills. Featured listings carousel.	Sidebar collapses to icon-only. No chat history stored or displayed. New Chat clears current session context in Redis.
Category browse pages	Four pages with left filter panel and listing grid. Shared filters: Location (multi-select), Verification status (default: Fully Verified + Doc Verified pre-checked), Amenities, Sort.	Filters apply without full page reload. Verification default can be overridden. Sort options functional.
Category-specific filters	Off-campus: institution (pre-filtered), gender (from profile, silent), unit type, available now toggle. Short-let: date range, guests, min stay, instant booking, min rating. Rent: bedroom count, type, furnishing, payment structure, available from, price range. Sales: bedroom count, type,	All filters apply correctly. Gender filter applied silently from seeker profile — not shown as a user-facing control. Date range filter excludes listings with any unavailability in selected period.

	development status, title type, price range.	
Map view toggle	Full-width Abuja pin map. Pins colour-coded: teal (fully verified), blue (doc verified), grey (unverified). Click-to-preview card overlay.	Map renders in under 2 seconds. All active filters apply to pin set. Pin click shows compact card without navigating away.
Listing page — core sections	Gallery (room-grouped, inspector walkthrough section separate). Title, subtitle, stars (not sales), badges, award badges. Host card (short-let only). Description with read-more. Amenities with confirmed indicators. Approximate location map.	Inspector walkthrough section labelled as BeeBop Verified. Confirmed amenity indicators show only for inspector-confirmed items. Approximate pin only — exact address not shown.
Listing page — fixed CTA bar	Type-specific: short-let date picker + dynamic price, student institution + semester, rent duration in years, sales no selector. Fixed to viewport bottom. Unauthenticated: Sign in to continue button.	CTA bar remains fixed during scroll. Price updates dynamically. Sign in button redirects to auth with return URL preserved.
Valuation report — gated	Section visible to all. Content (area scores, inspector note, report date) gated behind login. Locked panel shows description + registration prompt for logged-out users.	Locked panel describes report contents clearly. On login, report renders without page reload. Valuation report does not contain fair market value indicator or comparable listings.
Bookmark / save listing	Bookmark icon on card. Saved listings in seeker dashboard. Saved listings marked as unavailable when let/sold/delisted.	Bookmark state persists across sessions. Unavailable listings retain card with status label. No automatic removal.

7.4 Sprint 4 — Weeks 10–11: Admin Portal and Document Review

Goal: Admin can log into the internal portal, review document submissions, approve or query listings, and issue document badges.

Story	Detail	Acceptance Criteria
Admin authentication	Staff portal at internal route (e.g. /internal). Admin account seeded in database. FastAPI role guard enforced.	Internal route not linked from any public page. Admin role required for all /internal routes.
Document review queue	List of listings pending review. Sorted oldest first. Each entry: listing name, landlord name, submission date, document count, action buttons (Approve, Query, Reject).	Queue sorted by submission date. Approve triggers badge. Query opens note field and notifies landlord. Reject with reason notifies landlord.

Document viewer	In-portal PDF and image viewer using short-expiry S3 presigned URLs. No download required to review.	All document types render. URLs expire after 15 minutes. Regenerated on re-access.
Badge issuance — document badge	On approval: badge record created (admin ID, timestamp, 24-month expiry). Listing status updated. Landlord notified via email and WhatsApp.	Badge record created. Listing card updates. Landlord notification dispatched within 60 seconds.
Listing management	Admin view of all listings. Edit, suspend, soft-delete.	All admin edits logged with timestamp and admin ID. Suspended listings hidden from public. Soft-delete retains data.
Notification service — core events	Email via Resend for: offer accepted, agreement ready to sign, badge issued, agreement copy. WhatsApp via Business API for: new offer, offer accepted, visit confirmed, agreement ready to sign. In-app WebSocket for all events.	Emails sent via Resend. WhatsApp messages use approved templates. In-app notification delivered via WebSocket. All dispatches logged with status.
WhatsApp template integration	FastAPI service calls WhatsApp Business API for transactional messages using approved templates. Template parameters filled from event data.	Only approved templates used. Failed delivery logged. No fallback to unapproved free-form messages.
Celery worker setup	Celery workers running. Notification dispatch tasks defined. Beat scheduler configured for offer expiry timers.	Tasks execute asynchronously. Beat scheduler fires at correct intervals. Worker health monitored via Flower or equivalent.

7.5 Sprint 5 — Weeks 12–14: All Dashboards

Goal: Seeker, landlord (all types), student accommodation, and short-let dashboards complete and functional.

Dashboard	Key Stories
Seeker	Saved listings grid with unavailability status. Active offers list with response timer display. Agreements section with days remaining, PDF download, renewal prompt at 30 days.
Landlord — standard	Overview panel (active listings, pending actions badge, unread notifications). Listings section with all six status states. Offers and enquiries with Accept/Counter/Reject and 48-hour timer. Visits (read-only, no landlord coordination). Agreements with pending signature state. Fees and billing (facilitation fees itemised by transaction — no listing fees). Analytics (views, enquiry rate, avg time to offer).

Student accommodation	Inventory PMS: unit type panel with gender breakdown (F x N, M x N). Room-level grid with bed status (Occupied, Available, Vacating) and gender tag per room. Gender tag locked at database level when room occupied. Enquiries action per available bed. Waitlist with gender breakdown per unit type.
Short-let	14-day rolling availability calendar (scrollable to 90 days). Day states: Available, Booked, Turnover. Upcoming booking cards. Booking requests with Accept/Decline. Pricing settings panel (base rate, weekend rate, min stay, turnaround, instant booking toggle). Revenue and occupancy analytics.

Phase 1 Gate

Seekers can register (standard and student), browse all four categories with working filters, save listings, and view full listing pages including gated valuation report. Landlords can register, create all listing types for free, and manage via dashboard. Admins can review submissions and issue document badges. Email and WhatsApp notifications dispatch correctly. Zero critical or high-severity bugs on staging.

PHASE 2**Verification, Transactions and Messaging**Weeks 15 —
27 (13 weeks /
6–7 sprints)

Phase 2 delivers BeeBop's core differentiators: the physical verification pipeline, all transaction flows from offer to signed agreement with payment, and the short-let in-booking group chat with moderation.

8.1 Sprint 6 — Weeks 15–16: Inspector Portal and PWA

Story	Detail	Acceptance Criteria
Inspector onboarding flow	Admin creates inspector account. WhatsApp OTP invitation sent. Inspector activation: OTP, NIN verification, mandatory profile photo, area confirmation, conduct acknowledgement (standalone screen, explicit confirmation required).	Invitation link expires 48 hours. All steps mandatory. Inspector blocked from portal until complete.
Inspector assignment view	Inspector sees only properties assigned to them. Each entry: property name, area, type, briefing pack link.	No access to unassigned listings. Briefing pack mobile-accessible.
Property assessment form	Structured checklist: Identity/existence, Listing accuracy, Amenities (per listed item confirmed/not confirmed/absent), Structural condition. Free-text observation field. Progress saved to IndexedDB.	All fields required before submission enabled. IndexedDB saves on every field change.
Infrastructure scoring form	Road condition, NEPA/PHCN supply hours, security, proximity — each scored 1–5. Stored against GPS coordinates (area level), not listing ID. Submittable independently of assessment form.	Area-level storage confirmed — multiple listings sharing area share same score record. Independent submission works.
Photo and video capture with metadata	Camera access from mobile browser. GPS and timestamp captured per photo via browser Geolocation API and Date. Upload via presigned S3 URL with progress indicator.	GPS and timestamp stored in upload record (not reliant on EXIF). Upload continues in background after form submission initiated.
GPS location marking	Map interface for property pin placement. Device GPS auto-suggest on load. Manual adjustment available.	Coordinates stored to 6 decimal places. Auto-suggest requests geolocation permission correctly.
Report submission	Packages assessment, infrastructure scores, photos, video, GPS into atomic	Partial submissions rejected. Confirmation screen shows

	submission. Enters admin review queue.	submission reference. Inspector sees Pending status.
PWA offline capability	Service worker (next-pwa) caches form shell and static assets. IndexedDB stores all form data offline. Background sync dispatches queued submissions on reconnect. Sync status indicator visible at all times.	Form fully functional with no network. No silent sync failure — status always visible. Queued submissions dispatch in order on reconnect.

8.2 Sprint 7 — Weeks 17–18: Admin Inspection Review and Badge Pipeline

Story	Detail	Acceptance Criteria
Inspection report review queue	Admin views submitted reports. Each entry: inspector name, property, submission date, status. Open report shows checklist responses, infrastructure scores, photo/video gallery, GPS map.	Evidence renders in-portal. GPS shown on map. No download required.
Physical badge issuance	On approval: badge record created (inspector ID, admin ID, timestamp, 12-month expiry). Fully Verified check: if doc badge also present, Fully Verified designation applied. Landlord notified via email and WhatsApp.	Expiry date correct. Fully Verified fires only when both badges present. Notification within 60 seconds.
Area infrastructure score publication	Scores stored against GPS coordinates. Published to listing page as area score section. Admin can update scores at any time independently of badge status.	Scores display with last-updated timestamp. Update does not touch badge record. Multiple listings sharing coordinates share same score record.
Valuation report generation	On physical badge issuance: Claude API call generates report containing area infrastructure scores, inspector professional note, and report date. No fair market value indicator. No comparable listings. Report stored as structured JSON + formatted text against listing. Regenerated on area score update.	Report content confirmed: three sections only (area scores, inspector note, report date). Gated display works. Pydantic model validates Claude output before storage.
Confirmed amenity indicators	Inspector-confirmed amenities flagged in listing JSONB data on badge approval. Listing page renders Confirmed indicator per flagged item.	Confirmed indicators appear only for items inspector marked as present. Unconfirmed items show without indicator.
Valuation report — Redis caching	Report cached in Redis on generation. Cache invalidated on area score update	Cache hit confirmed in monitoring. Cache miss triggers regeneration

	or listing price change. Not regenerated on every page view.	and re-caches. No stale report served after invalidating event.
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8.3 Sprint 8 — Weeks 19–20: Offer and Counter-Offer Flow

Story	Detail	Acceptance Criteria
Offer submission	Post-visit offer (unlocked after admin approves visit report) and pre-visit direct offer. Fields: offered price, move-in date, conditions. Landlord notified via email and WhatsApp.	Offer record created with all fields. Post-visit offer gated at API level until visit report approved. Notification within 60 seconds.
Offer response — landlord	Accept, Counter (opens price input), or Reject. 48-hour response window. Celery task schedules staged reminder notifications.	Timer starts at receipt. Acceptance triggers agreement generation pipeline. Counter creates linked counter-offer record. Rejection triggers in-app notification to seeker only.
Counter-offer loop	Maximum 3 rounds enforced at API level. Session expires on round 4 attempt or 48-hour non-response.	API rejects fourth counter with clear error. Expiry notifies both parties. Listing reopens to other seekers on expiry.
Staged offer expiry notifications	Celery Beat: Hour 0 email+WhatsApp, Hour 24 in-app only, Hour 36 email+WhatsApp with time remaining, Hour 48 expiry email+WhatsApp both parties.	All Celery tasks scheduled correctly on offer creation. Tasks cancelled if offer actioned before timer fires.
Visit request creation	On offer acceptance where no visit has been conducted: visit request auto-created in admin visit queue.	Visit request record created linking listing ID, seeker ID, timestamp. Admin sees new queue entry.
Admin manual visit assignment	Admin selects agent from available agents list in portal. Assignment stored. Agent notified via WhatsApp and in-app. 2-hour confirmation window.	No automated assignment engine. Admin selects manually. Unconfirmed assignments alert admin at 2-hour mark.

8.4 Sprint 9 — Weeks 21–22: Agent Workflow and Visit Management

Story	Detail	Acceptance Criteria
Trusted agent onboarding	Admin creates account. WhatsApp OTP invitation. Activation: OTP, NIN verification, mandatory profile photo, area confirmation, conduct acknowledgement.	All steps mandatory. Conduct acknowledgement is standalone screen — not buried in terms. Agent blocked from portal until complete.

Agent portal — visit schedule	Confirmed visits sorted by date: property name, address, GPS, date, time, seeker first name, briefing pack. Past visits archived.	Mobile-accessible. No seeker contact details shown. Briefing pack loads without page reload.
Post-visit report — agent	Visit confirmation section (did visit occur, access issues, conduct issues) and property observations section (amenity checklist vs listing, discrepancies, free-text).	Submission atomic. Linked to visit, agent, and listing records. Enters admin review queue.
Admin visit report review	Approve: unlocks seeker post-visit offer. Query: returns to agent with note. Flag: creates listing review action.	Post-visit offer unlocked only on admin approval. Seeker notified via in-app and WhatsApp within 60 seconds of approval.
Visit cancellation	Either party can cancel via BeeBop. Cancellation record created with reason. Agent notified immediately. Listing reopens for new visit requests.	All parties notified on cancellation. Agent WhatsApp notification. Visit record updated — not deleted.
Role separation enforcement	System prevents same person being inspector and visit agent for the same listing. Enforced at assignment time in admin portal.	Assignment UI shows conflict warning if agent was inspector for that listing. API-level check also blocks the assignment.

8.5 Sprint 10 — Weeks 23–25: Agreement Generation, Signing and Fees

Note: Agreement templates must be signed off by the property lawyer before this sprint begins.

Sprint cannot start on agreements until templates are ready.

Story	Detail	Acceptance Criteria
Tenancy agreement generation	On offer acceptance: FastAPI generates tenancy agreement (rent) or sale memorandum (sales) pre-filled from platform data. Stored as PDF (Weasyprint or ReportLab) in S3 and as structured data in database.	Two templates: rent and sale. Generated within 30 seconds of offer acceptance. Draft sent to both parties for review.
OTP-confirmed signing	Both parties review agreement in dashboard. Sign via OTP on registered email or WhatsApp. Both signatures required. Signing event timestamped.	Partial signature state visible. Agreement marked Signed only when both OTPs confirmed. Signed PDF regenerated with timestamps.
Agreement storage and download	Permanently stored in S3. Downloadable by both parties at any time via short-expiry presigned URL.	Download available from both dashboards. URL regenerated on each request.
Rent facilitation fee — Paystack	On signing: landlord charged 15,000 Naira + 2% (standard) or 1.34%	Tier threshold check correct at 5,000,000 Naira. Both Paystack

	(premium) of annual rent. Seeker charged 5,000 Naira + 1% (standard) or 0.66% (premium). Tier determined by agreed annual rent. Both charged via Paystack.	charges created. Payment failure holds agreement in pending-payment state. Both parties see clear payment status.
Student accommodation fee — Paystack	Accommodation owner charged 2.5% of agreed price. Seeker charged 3% of agreed price. Independently charged.	Charges calculated on agreed accommodation price for the agreed term. Two separate Paystack payment records.
Sales fee — Paystack invoice to seller	On agreement signing: Paystack invoice generated to seller for 3% (standard) or 2.5% (premium) of sale value. Tier at 50,000,000 Naira. Signed agreement withheld until invoice paid. 48-hour payment window.	Invoice sent to seller's email. Agreement PDF released to both parties on payment confirmation. If unpaid after 48 hours: admin alert generated. Buyer not charged anything.
Listing status update	Listing marked Let agreed (rent/student) or Sale agreed (sales) within 60 seconds of final signature confirmation.	Status update reflected on listing card platform-wide. Listing visible but clearly marked.
Renewal fee — Paystack	30 days before expiry: landlord prompted to renew. Renewal triggers simplified agreement flow. Landlord charged 10,000 Naira renewal fee via Paystack on execution.	Renewal fee charged to landlord only. Simplified flow reuses existing listing data. New agreement generated with updated dates.

8.6 Sprint 11 — Weeks 25–26: Short-Let Booking and Payment

Story	Detail	Acceptance Criteria
Instant booking flow	Date selection on CTA. Dynamic price calculation (base × nights + weekend uplift). Paystack payment. Booking confirmed on success. Access details released on check-in day via email and WhatsApp.	Weekend rate applied correctly. No booking created on payment failure. Access details not released before check-in day.
Booking request flow	Request submitted to host. Accept/Decline in host dashboard. Payment captured on acceptance only. Seeker notified of decision via email and WhatsApp within 24 hours.	Payment not taken on request. Decision notification within 24 hours enforced via Celery task.
Seeker fee calculation	At booking creation: if total booking value above 150,000 Naira, seeker fee = 8% of booking value. At or below: seeker fee = 10%. Added to booking total at checkout.	Threshold check at booking value (total stay). Fee displayed clearly before payment confirmation.
Host payout	After check-in confirmation: Paystack transfer to host bank account. Host	Payout record created. Host receives WhatsApp confirmation.

	payout = gross booking value minus 3% host fee.	Net payout calculation confirmed correct.
Availability enforcement	Booked dates blocked. Turnaround days auto-blocked after check-out based on host setting. API-level double-booking prevention.	Concurrent booking attempts handled transactionally. No double-booking possible. Turnaround days calculated from check-out date.

8.7 Sprint 12 — Week 27: Short-Let In-Booking Group Chat

Goal: Every confirmed short-let booking has an associated group chat thread. Seeker, host, and BeeBop moderator can communicate within the platform. Content filtering flags attempts to move off-platform.

Story	Detail	Acceptance Criteria
Chat thread creation	On short-let booking confirmation: chat thread created automatically. Participants: seeker user, host user, BeeBop moderator account. Thread linked to booking record.	Thread created within 5 seconds of booking confirmation. All three participants have access. Thread cannot be created without a confirmed booking.
Real-time messaging — WebSocket	Chat UI built in Next.js using native WebSocket. Messages sent to FastAPI WebSocket endpoint. Broadcast to all thread participants. Message stored in database.	Message delivered to all active participants in under 1 second. Participant who is offline sees messages on next connection. No messages lost.
Chat UI — booking card integration	Chat accessible from booking card in seeker dashboard and host booking management. Opens as panel or modal — no separate page navigation required.	Chat opens within the dashboard context. Previous messages load on open (last 50 messages). Input field focused automatically.
On-platform action shortcuts	When seeker asks about extending stay, system detects extension intent and renders an Extend booking inline action button within the chat. Other shortcuts: Report issue, Request early check-in.	Action buttons rendered inline in chat. Clicking opens the relevant platform flow (e.g. date extension picker) — not a WhatsApp redirect.
Content filtering — pattern detection	Python regex patterns for: Nigerian phone number formats (080x, 090x, 070x, 081x, 091x, 071x), account numbers (10-digit strings), bank names (list of major Nigerian banks), common off-platform payment phrases.	Flagged messages not blocked — delivered but marked internally as flagged. Flagged message triggers moderator alert in admin portal.
Moderator alert queue	Admin portal shows flagged messages with full context. Moderator can: dismiss	Alert appears in moderator queue within 30 seconds of flag. Warning

	(false positive), warn participant (sends template warning in chat), or escalate (suspend participant account pending review).	message sent as BeeBop system message in chat. Escalation creates admin action record.
Chat moderation UI	Moderator sees all chat threads grouped by booking. Can read all messages. Can send messages as BeeBop. Can flag, warn, or escalate.	Moderator view clearly separated from participant view. Moderator messages displayed with BeeBop label.
No chat for other listing types	Chat feature is strictly short-let only. Rent, sale, and student accommodation do not have chat threads.	Chat thread creation blocked at API level for non-short-let bookings.

Phase 2 Gate

Full verification pipeline operational end-to-end. Inspector portal with offline capability live. All transaction flows (rent, student, sales, short-let) complete with payment. In-booking group chat with moderation live. All dashboards complete. Zero critical or high-severity bugs on staging. Property lawyer agreement templates signed off and integrated.

PHASE 3	Beta Launch	Weeks 28 — 37 (10 weeks)
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Phase 3 takes the platform from technically complete to a product in the hands of real users. A controlled beta surfaces usability issues that internal testing cannot find and produces the first verified listings demonstrating BeeBop's trust model in practice.

9.1 Sprint 13 — Weeks 28–29: AI/NLP Layer

Story	Detail	Acceptance Criteria
Intent classification	Claude API integration via Anthropic Python SDK. System prompt includes: Nigerian housing vocabulary map, Abuja location synonym list, intent rules. Four intents: search, clarification, information, transactional.	95%+ correct classification on pre-built 200-query test set.
Parameter extraction — Pydantic validation	Claude returns structured JSON. Pydantic model validates: property_type, listing_category, location, budget, amenities, verification_level, duration, urgency. Validation failure triggers retry (max 2). Fallback to PostgreSQL FTS on retry exhaustion.	Valid Pydantic model returned for 98%+ of well-formed queries. Fallback fires silently. All events logged.
Session context — current session only	Session context (current search parameters, last result set as listing IDs, last 10 conversation turns) stored in Redis with session-scoped key. TTL: 30 minutes inactivity. Cleared on New Chat. No database persistence.	Context correctly retrieved across turns within session. Context expires correctly. New Chat clears Redis key immediately. No cross-session context leak.
Reference resolution	References in subsequent turns (ones, the third one, what about X) resolved against current session context.	Pronoun and ordinal references correctly resolved against last result set in 90%+ of test cases.
Results window — three states	Collapsed (first few results), expanded (full grid), minimised (session input only, no results). Pagination for 100+ results.	State transitions smooth. Pagination correct. Session input visible in all states.
Result ranking — updated model	Signals: (1) Verification status — highest weight. (2) Rating score — second weight. (3) Parameter match precision. (4) Recency. (5) Price competitiveness — light signal. Saves and enquiries not used.	Ranking order correct in manual evaluation against 50 test scenarios. Rating signal confirmed: high-rated listing ranks above unrated listing of same verification tier.

Query logging	Every query: raw text, classified intent, extracted parameters, returned listing IDs, session ID, timestamp. Click-through events logged against query ID.	Log entry created for every query. Click-through event recorded when seeker opens a returned listing.
Fallback to keyword search	If Claude fails after retries: PostgreSQL FTS query using raw input terms. Seeker sees results without error.	Fallback fires correctly. Fallback event logged. Seeker experience uninterrupted.

9.2 Beta Preparation — Weeks 30–33

Area	Activities
Performance	k6 load test on staging: 500 concurrent users, 1,000 listing page views/min, 100 concurrent AI queries. All image delivery via Cloudinary with responsive sizing. EXPLAIN ANALYSE on all PostgreSQL queries touching listings and users tables. Redis caching for listing cards and area scores.
Security audit	Engage security consultant. Scope: FastAPI auth flows, API authorisation, file upload validation, SQL injection/XSS, S3 bucket policies, secrets exposure, WebSocket auth. All critical and high findings resolved before beta opens.
Beta listings	Recruit 50+ real Abuja landlords across all four categories (minimum: 10 off-campus, 10 short-let, 20 rent, 10 sales). All doc submissions processed before beta opens. Physical inspections conducted for as many listings as possible.
Beta seekers	200–300 invited seekers from Abuja networks (university notice boards, housing WhatsApp/Facebook groups, personal networks). Staggered invitations. In-app feedback button built into beta build.
Upgrade Render to paid plan	Move from Render free tier to Render Starter (\$7/month) before beta opens. At this point real users are on the platform and cold starts are unacceptable. Disable the keepalive workflow once Render is on a paid plan.
Legal and compliance	NDPR compliance review completed. Privacy policy and terms of service published. Paystack compliance review done. WhatsApp Business API templates all approved. Sales invoice process legally reviewed.
Support infrastructure	Support email and WhatsApp business number live. Internal knowledge base for support team. Escalation path to admin defined.

9.3 Beta Live and Iteration — Weeks 34–37

Week	Activity
Week 34	Beta opens to first 100 seeker invites. All admin, inspector, agent, and moderator staff operational. Daily monitoring: error rates, notification delivery rates, offer response rates, chat moderation volume, support ticket volume.

Week 35	First structured feedback session with beta landlords and seekers. Findings triaged: P1 (fix within week), P2 (fix before public launch), P3 (backlog). P1 fixes deployed to staging and promoted to production.
Week 36	Second cohort (remaining seekers) invited. Performance metrics reviewed — scaling actions if degradation observed. First fully verified listings (both badges) published.
Week 37	Final beta synthesis. Go/no-go assessment by stakeholders. All P1 and P2 issues resolved. Phase 4 sprint backlog populated. Public launch date confirmed.

Phase 3 Gate

Minimum 50 real listings live. At least 10 with document badge, at least 5 fully verified. AI/NLP processing real queries with measurable accuracy. Security audit findings resolved. All legal and compliance requirements met. Go/no-go decision made.

PHASE 4

Production Launch and Optimisation

Weeks 38 —
52 (15 weeks)

Phase 4 is public launch and systematic optimisation driven by real usage data. Primary build work covers the rating and review system, area knowledge base infrastructure, and NLP vocabulary improvement.

10.1 Public Launch — Week 38

Activity	Detail
Staged rollout	General public access. Alert thresholds: error rate above 1%, notification delivery failure above 5%, listing page load above 3 seconds, API p95 latency above 500ms.
Marketing	Launch announcement. University student union partnerships for off-campus category. Targeted outreach in Abuja housing communities via WhatsApp and social media.
Monitoring cadence	Daily engineering and ops review for first two weeks. Defined incident escalation protocol in place.

10.2 Rating and Review System — Weeks 38–42

Story	Detail	Acceptance Criteria
Eligibility gating	Review prompt triggered only for verified transactors. Rent/student: 30 days after agreement signing. Short-let: 24 hours after check-out. Via in-app notification and email.	Prompt not shown to browsers. No review possible without completed transaction.
Short-let review form	Overall star rating (1–5) + four sub-category ratings: Accuracy, Cleanliness, Location, Value. Optional written review (min 50 chars if provided).	Overall rating required. Sub-categories optional. Written review requires overall rating to be set first.
Rent and student review form	Overall star rating + optional written review.	Same validation pattern.
Review display	Reviews carousel at bottom of listing page. Newest first. Short-let cards show sub-category breakdown. Sub-category averages displayed.	Pagination for listings with more than 10 reviews. Sub-category averages calculated from all qualifying reviews, not just those with written content.

Award badge automation	Guest Favourite: short-let with 4.5+ average from 10+ reviews in last 12 months. Highly Responsive: landlord with 90%+ offer response rate within 24 hours over last 30 days. Nightly Celery Beat job. Revoked when threshold no longer met.	Nightly job runs correctly. Badge appears on listing card within 24 hours of threshold crossing. Revocation fires correctly.
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10.3 NLP Optimisation — Weeks 39–46

Area	Activity	Metric
Vocabulary expansion	Review query logs for terms producing zero results or wrong intent. Add to vocabulary map. Re-test affected patterns.	Zero-result rate for well-formed queries below 5%.
Location synonyms	Identify unmatched location references in query logs. Add estate names, colloquial names, landmarks to Python synonym dict.	Location extraction accuracy above 95%.
Parameter extraction precision	Analyse parameter mismatches. Refine Claude prompt for edge cases. Version all prompt changes.	Precision above 90% on 500-query labelled test set.
Prompt version control	All changes committed to prompt registry module. A/B testing via PostHog feature flags on live traffic.	No prompt change in production without staging evaluation period.

10.4 Knowledge Base Infrastructure — Weeks 42–50

Story	Detail
Area score aggregation	Scores normalised per geographic grid cell. FastAPI endpoint to query by coordinates or district name. Foundation for area intelligence pages.
Price history tracking	Every listing price change creates a history record. Queryable by listing, area, and property type. Feeds future valuation report improvements.
Area intelligence pages	Public-facing district pages showing aggregated infrastructure scores. Updated from area score data. First published intelligence product.
Valuation report improvement	Expand with price trend context from price history. Improve area score display with trend direction (improving / stable / declining).

10.5 Week 52 Success Metrics

Metric	Target
Live verified listings (both badges)	200+
Registered seekers	3,000+
Registered landlords / agents	500+
Completed rent transactions	100+
Completed short-let bookings	300+
AI query zero-result rate	Below 5%
Notification delivery rate (email + WhatsApp combined)	Above 97%
Platform uptime	Above 99.5%
API p95 latency	Below 400ms
Area infrastructure data points	500+ unique property assessments
Short-let chat moderation — off-platform attempt rate	Tracked and trending down month-over-month

11. Quality Assurance Strategy

QA runs in parallel with every sprint from Phase 1 onwards. The QA engineer is embedded in the team, not a downstream function.

11.1 Testing Layers

Layer	Coverage	Tooling
Unit tests	All business logic: fee calculation (all tiers and thresholds), badge eligibility rules, offer expiry timer scheduling, parameter extraction validation, notification dispatch conditions, chat content filtering pattern matching.	pytest
Integration tests	FastAPI endpoint contracts: auth flows, listing CRUD, offer state machine transitions, Paystack payment integration, notification dispatch, chat thread creation and messaging.	pytest + httpx (async test client)
End-to-end tests	Critical journeys: seeker registration and first search, landlord listing creation (free), offer to agreement signing, short-let booking flow, inspector report submission (including offline/sync).	Playwright
NLP test suite	500 labelled queries with expected intent and parameter outputs. Runs on every prompt change in prompt registry. Pydantic schema validation tested separately.	Custom pytest suite against Anthropic API (test account)
Performance tests	500 concurrent users. 1,000 listing page views/min. 100 concurrent AI queries. 50 concurrent WebSocket chat connections.	k6
Security tests	OWASP Top 10. Auth bypass. File upload validation. API authorisation boundaries. WebSocket message injection. Chat content filter bypass attempts.	Manual (consultant) + OWASP ZAP

11.2 Bug Severity and Resolution SLAs

Severity	Definition	SLA
Critical (P0)	Data loss, security breach, payment failure, auth broken, production outage, double-booking possible.	Fix and deploy within 4 hours.

High (P1)	Core user journey broken. Badge not issuing. Notification not dispatching. Chat flagging not working. Fee calculation error.	Fix within 24 hours. Same-day production deploy.
Medium (P2)	Non-critical feature broken. Filter not applying correctly. Dashboard display issue.	Fix within current sprint.
Low (P3)	Minor UI inconsistency. Copy error. Edge case in non-critical flow.	Next available sprint.

All P0 and P1 fixes require a regression test added to the automated suite before the fix is merged.
No exceptions.

12. Risk Register

Risk	Likelihood	Impact	Mitigation
WhatsApp Business API template approval delayed	Medium	High	Submit all message templates (OTP, offer notification, badge issued, visit confirmed, agreement ready) in Phase 0 immediately. Templates take 24–72 hours to approve but can be resubmitted if rejected. Email OTP via Resend as fallback during approval period.
NIMC API access delayed	Medium	High	Initiate application in Phase 0 immediately. Build manual NIN verification fallback (admin reviews uploaded NIN document) to operate if API unavailable at launch. Monitor NIMC developer portal status.
Paystack onboarding rejected or delayed	Low	High	Apply in Phase 0. Ensure all KYC documentation prepared. Flutterwave as backup payment gateway — same Paystack-like API surface.
Sales invoice payment not received after signing	Medium	Medium	48-hour payment window enforced by Celery task. Admin alert on non-payment. Agreement released after 72 hours with admin review to prevent indefinite blocking of legitimate transactions.
Short-let chat — sustained off-platform circumvention	Medium	High	Content filtering provides detection. Moderator action (warn, then suspend) provides deterrent. In-platform booking shortcuts reduce motivation to go off-platform. Review circumvention rate monthly and tighten patterns as evasion tactics emerge.
Python backend developer availability	Low	Medium	Python is widely available globally and increasingly in Nigeria. FastAPI is a mainstream framework. Documented architecture reduces single-developer dependency.
Inspector capacity insufficient for beta	Medium	Medium	Recruit minimum 3 inspectors before beta. Batch by area. Physical badge not required for listing eligibility — doc badge sufficient to go live. Inspections continue in parallel.
Celery task failures causing missed notifications	Low	High	Celery worker health monitored via Flower. Dead letter queue for failed tasks. Retry policy (max 3 retries with exponential backoff) on all notification tasks. Admin alert on worker failure.
NDPR compliance challenge	Low	High	Engage legal counsel in Phase 0. Privacy policy published before beta. NIN and BVN data not retained — only verified/unverified status stored.
Key developer departure	Low	High	All architecture decisions documented in repo. No single-developer knowledge dependencies. Codebase bus factor reviewed in Phase 3.

Free tier service limits hit unexpectedly	Low	Medium	All free tiers have defined limits documented in Section 3.5. Monitoring via Sentry and PostHog catches approaching limits. Each service has a clear and affordable upgrade path. Render upgrade to \$7/month is the most likely and earliest upgrade needed.
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13. Master Timeline

Phase	Weeks	Duration	Key Deliverable
Phase 0 — Foundation	1–3	3 weeks	Infrastructure, Python + Next.js projects, WhatsApp Business API, schema, CI/CD.
Phase 1 — Core Platform MVP	4–14	11 weeks	Free listing creation, browse, listing page, admin doc review, all dashboards.
Phase 1 Gate Review	14	3 days	QA sign-off. Staging validated. Stakeholder review.
Phase 2 — Verification, Transactions and Messaging	15–27	13 weeks	Inspector PWA, verification pipeline, full transaction flow, payments, short-let chat.
Phase 2 Gate Review	27	3 days	QA sign-off. Legal templates integrated. Security audit initiated.
Phase 3 — Beta Launch	28–37	10 weeks	AI/NLP layer, beta listings, beta seekers, go/no-go decision.
Phase 3 Gate Review	37	3 days	Go/no-go assessment. All P1+ issues resolved.
Phase 4 — Production Launch	38–52	15 weeks	Public launch, rating system, knowledge base, NLP optimisation, scaling.
Week 52 Review	52	1 week	Success metric review. Phase 5 planning.

Total Programme Duration

12 months from Phase 0 kick-off to Week 52 review. Public launch target: Week 38 (approximately Month 9.5). Revenue-generating features — short-let bookings, rent and student facilitation fees, and sales commissions — are all live before public launch.

14. Open Prerequisites Before Development Begins

These items involve external parties with lead times outside the team's control. Every day of delay in Week 1 directly risks the programme timeline.

Item	Action	Owner	Deadline
WhatsApp Business API	Create Meta Business Manager account. Apply for WhatsApp Business API access. Submit all message templates for approval simultaneously.	Tech Lead + WhatsApp Specialist	Week 1, Day 1
NIMC API access	Submit developer application. Provide required business documentation.	Founder / Tech Lead	Week 1
CAC API access	Submit application for CAC registration verification API access.	Founder / Tech Lead	Week 1
Paystack merchant account	Create and verify. Complete KYC. Configure settlement account. Obtain test and live API keys.	Founder	Week 1
Property lawyer engagement	Retain lawyer to draft tenancy agreement and sale memorandum templates. Both required by Phase 2 Sprint 10 (Week 23).	Founder	Week 1–2
NDPR compliance review	Engage legal counsel. Privacy policy and data handling documentation required before beta.	Founder / Legal	Week 2
beebop.ng domain	Purchase confirmed. Register via NiRegistry or reseller. Point DNS nameservers to Vercel for frontend hosting. Configure subdomain for Render backend (api.beebop.ng).	Tech Lead / Founder	Week 1
Google Maps Platform	Create GCP project. Enable Maps JavaScript API, Geocoding API, Places API. Set billing with budget alerts. \$200/month free credit covers MVP usage.	Tech Lead	Week 1
Hosting accounts (Vercel, Render, Neon, Upstash, Cloudinary)	Create all five free-tier accounts. Connect GitHub repos to Vercel and Render. Obtain connection strings and API keys. Store as environment variables.	Tech Lead	Week 1
Anthropic and OpenAI API keys	Accounts already funded. Obtain production API keys. Set monthly	Tech Lead	Week 1

	budget alerts on both accounts. Store as environment variables in Render and Vercel dashboards.		
Resend email domain	Create Resend account. Verify beebop.ng sending domain. Configure DKIM/SPF records on domain DNS. Test OTP email template delivery.	Tech Lead	Week 1
Inspector recruitment	Identify and vet 3 initial inspectors from surveying firm. Must be onboarded before beta (Week 28).	Founder / Operations	By Week 12
Trusted agent recruitment	Identify and vet 3 trusted agents for Abuja. Must be onboarded before beta (Week 28).	Founder / Operations	By Week 12
Short-let chat moderator	Identify who will moderate short-let chats at launch. Can overlap with platform admin role at low volume.	Founder / Operations	By Week 20

15. Glossary

Term	Definition
Alembic	Python database migration tool. Works with SQLAlchemy to track and apply schema changes as versioned scripts.
Anthropic Python SDK	Official Python library for calling the Claude API. Used directly without LangChain or similar abstraction.
CAC	Corporate Affairs Commission. Nigeria's business registration authority.
Celery	Python distributed task queue. Handles async background jobs: notifications, agreement generation, badge awards, expiry timers. Redis as broker.
FastAPI	Python web framework. Async-first, high performance, native Pydantic integration, automatic OpenAPI documentation.
IndexedDB	Browser-native key-value database used by the inspector PWA to store form data offline before background sync.
Meta Business Manager	Meta's platform for managing WhatsApp Business API access, message templates, and business verification.
NDPR	Nigeria Data Protection Regulation. Primary data privacy law governing personal data in Nigeria.
NIMC	National Identity Management Commission. Issues NINs and provides the API for real-time NIN verification.
NIN	National Identification Number. 11-digit identifier issued to Nigerian citizens and residents by NIMC.
Paystack	Primary payment gateway. Handles all BeeBop payment events including card payments, bank transfers, invoicing, and payouts.
Pydantic	Python data validation library. Used for API request/response models and LLM output schema validation.
PWA	Progressive Web Application. Browser-based application with offline capability, installability, and background sync via service workers.
Redis	In-memory data store. Used for session context (current conversation), Celery task queue, valuation report cache, and listing card cache.
Resend	Transactional email delivery service. Handles OTP emails, agreement PDFs, fee receipts, and onboarding confirmations.
SQLAlchemy	Python ORM. Provides type-safe async database access. Used with Alembic for migrations.
WeasyPrint / ReportLab	Python PDF generation libraries. Used to produce tenancy agreement and sale memorandum PDFs at signing.
UptimeRobot	Free uptime monitoring service. Pings the /health endpoint every 5 minutes to prevent Render free tier spin-down and to alert on genuine downtime.
WhatsApp Business API	Meta's programmatic API for sending WhatsApp messages using approved templates. Used for OTP verification and time-sensitive notifications.

